

# TLAY Physical Energy Service

API Specification | 0.2.0-review | 27 September 2026

## Scope and status

Application contract for discovering, purchasing and tracking a fixed-duration eCandle power session. One paid endpoint; five supporting business/identity endpoints; one health endpoint. Integration review version: deployment and end-to-end validation pending.

Base URL: supplied by the operator at deployment. All paths below are relative. The complete OpenAPI 3.0.3 definition follows this reference summary and is also provided as openapi.yaml.

## Endpoint reference

| Method | Path                                     | Charge          |
|--------|------------------------------------------|-----------------|
| GET    | /v1/energy/offers                        | Included / free |
| POST   | /v1/energy/quotes                        | Included / free |
| POST   | /v1/energy/sessions                      | Paid            |
| GET    | /v1/energy/sessions/{session_id}         | Included / free |
| GET    | /v1/energy/sessions/{session_id}/receipt | Included / free |
| GET    | /v1/devices/{device_id}/keys/{key_id}    | Included / free |
| GET    | /healthz                                 | Included / free |

## Purchase contract

POST /v1/energy/sessions accepts quote\_id and client\_nonce. Idempotency-Key is required. Unpaid calls return 402 with PAYMENT-REQUIRED; the SDK-generated paid retry uses PAYMENT-SIGNATURE. 202 returns session and access\_token. It confirms order acceptance, not completed power delivery.

# Request, response and lifecycle

## Illustrative purchase request

```
{
  "quote_id": "quote_demo",
  "client_nonce": "buyer_nonce_demo"
}
```

## Principal objects

| Object         | Required meaning                                                                       |
|----------------|----------------------------------------------------------------------------------------|
| Offer          | Authorized site/output, duration, maximum load, decimal USDC price, network and terms. |
| Quote          | Immutable offer and request digest, expiry, start deadline and refund rule.            |
| Session        | Execution state, payment state, delivery counters and payment reference.               |
| Receipt        | Order association, signed measurements, terminal delivery state and proof status.      |
| DeviceIdentity | Versioned public key, signature profile and key validity/revocation status.            |
| ProofBundle    | Versioned verification profile and portable artifact.                                  |

## Execution and proof

accepted -> starting -> running -> completed. Failed and partially\_completed are terminal alternatives. Receipt retrieval returns 202 while active and 200 for a terminal record. proof\_state is independently pending, ready or unavailable; ready requires proof\_bundle.

## Authorization and retry behavior

Site/output authorization uses access\_grant in quote creation. Status and receipt reads require a scoped bearer token. Payment authorization is protocol-specific. An authenticated idempotent replay must recover the original order without a duplicate charge or execution; an idempotency key alone cannot authorize access.

## Errors and financial state

400 invalid input; 401 missing/invalid token; 403 access/payment denial; 409 expired quote or conflict; 429 rate limit; 503 temporary failure. Reconcile ambiguous payment outcomes before initiating another purchase. Refunds follow the disclosed quote policy and are recorded separately.

# Verification and protocol boundaries

Device signatures cover exact measurement bytes. Decode payload\_b64, compare SHA-256, resolve the trusted key and validity interval, verify the allowlisted signature profile, then compare decoded device/session/time/measurement fields with the receipt. HashAnchor artifacts must bind the relevant request, measurement and payment records.

Payment acceptance, batch settlement, physical execution and proof availability are separate events. settlement\_tx\_hash is optional until actually known. Signature validity does not establish meter accuracy; operator measurement and acceptance conditions remain necessary.

## Deployment configuration

Confirm the production/test base URL, facilitator and SDK versions, network and asset identifiers, payTo address, prices, signing payload and algorithm, HashAnchor proof profile, receipt retention, authenticated recovery and refund process. The protocol payment payload is generated by the selected x402 SDK; this application schema does not redefine it.

## Normative OpenAPI definition

The following listing contains the full proposed application contract. All examples use illustrative IDs. Line wrapping in this PDF is for readability; use the accompanying YAML for tooling. Schema validation establishes structural validity, not implementation conformance.

```
openapi: 3.0.3
info:
  title: TLAY Physical Energy Service
  version: 0.2.0-review
  description: 'Application API contract for an eCandle physical energy service: offer discovery,
    authorized
    purchase, asynchronous execution and delivery receipts. Integration review version; deployment and
    end-to-end validation pending. Relative paths require an operator-provided base URL.'
  contact:
    name: TLAY
    url: https://www.tlay.io/
  x-implementation-status: proposed; deployment and interoperability tests pending
paths:
  /v1/energy/offers:
    get:
      operationId: listEnergyOffers
      summary: Discover available energy offers
      security: []
      responses:
        '200':
          description: Public offers. Availability is indicative until a quote reserves capacity.
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/Offer'
        '429':
          description: Rate limit exceeded.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
  /v1/energy/quotes:
    post:
      operationId: createEnergyQuote
      summary: Authorize and reserve a short-lived offer before payment
      security: []
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/QuoteRequest'
      responses:
        '201':
          description: Quote created. Same grant and nonce return the same unexpired reservation.
          content:
```

```

    application/json:
      schema:
        $ref: '#/components/schemas/Quote'
  '400':
    description: Invalid parameters.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  '403':
    description: Site/output permission denied.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  '409':
    description: Unavailable capacity or nonce conflict.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  '429':
    description: Rate limit exceeded.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
/v1/energy/sessions:
  post:
    operationId: purchaseEnergySession
    summary: Purchase one physical energy session via x402
    description: x402-gated purchase. An unpaid request receives 402. Retry the identical body and
    Idempotency-Key
      with a valid SDK-generated payment payload. 202 confirms an accepted session, not completed
    delivery.
      Payment network, asset, seller address and facilitator configuration are supplied by the
    deployment.
    security: []
    x-payment-protocol: x402; deployment profile pending confirmation
    parameters:
      - name: Idempotency-Key
        in: header
        required: true
        schema:
          type: string
          minLength: 16
          maxLength: 128
      - name: PAYMENT-SIGNATURE
        in: header
        required: false
        schema:
          type: string
        description: Protocol-generated payment payload on the paid retry; use the selected x402
  client
    SDK.
  requestBody:
    required: true
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/PurchaseRequest'
  responses:
    '202':
      description: Task accepted or an authenticated idempotent replay.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/AcceptedPurchase'
      headers:
        Location:
          description: Session status path.
          schema:
            type: string
    '400':
      description: Invalid request.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    '402':
      description: Payment requirements provided by the configured x402 SDK in PAYMENT-REQUIRED.
      headers:
        PAYMENT-REQUIRED:
          required: true
          description: SDK-generated, encoded x402 payment requirements.
          schema:

```

```

        type: string
      '403':
        description: Invalid authorization or payment.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
      '409':
        description: Expired quote, consumed reservation, mismatched request, or idempotency
        conflict:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
      '429':
        description: Rate limited.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
      '503':
        description: Temporarily unavailable. A payment already accepted must be reconciled before a
        new attempt.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
/v1/energy/sessions/{session_id}:
get:
  operationId: getEnergySession
  summary: Read execution and payment states
  security:
    - SessionToken: []
  parameters:
    - name: session_id
      in: path
      required: true
      schema:
        type: string
  responses:
    '200':
      description: Current session.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Session'
    '401':
      description: Missing or invalid token.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    '403':
      description: Token not authorized for this session.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
    '404':
      description: Session not found.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Error'
/v1/energy/sessions/{session_id}/receipt:
get:
  operationId: getEnergyReceipt
  summary: Retrieve a delivery receipt without another payment
  security:
    - SessionToken: []
  parameters:
    - name: session_id
      in: path
      required: true
      schema:
        type: string
  responses:
    '200':
      description: Terminal delivery receipt. proof_state may remain pending.
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/Receipt'
    '202':
      description: Physical session is not terminal yet.
      content:

```

```

    application/json:
      schema:
        $ref: '#/components/schemas/Session'
  '401':
    description: Missing or invalid token.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  '403':
    description: Access denied.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  '404':
    description: Session not found.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
/v1/devices/{device_id}/keys/{key_id}:
  get:
    operationId: getDeviceKey
    summary: Read versioned device identity and key status
    security: []
    parameters:
      - name: device_id
        in: path
        required: true
        schema:
          type: string
      - name: key_id
        in: path
        required: true
        schema:
          type: string
    responses:
      '200':
        description: Key metadata. Trust in the registration authority remains necessary.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/DeviceIdentity'
      '404':
        description: Unknown device or key.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Error'
/healthz:
  get:
    operationId: getServiceHealth
    summary: Read service health
    security: []
    responses:
      '200':
        description: Service health; does not promise physical capacity.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Health'
      '503':
        description: Degraded health.
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Health'
components:
  securitySchemes:
    SessionToken:
      type: http
      scheme: bearer
      description: Opaque, session-scoped token issued after purchase. This is separate from x402
      payment
      authorization.
  schemas:
    Offer:
      type: object
      additionalProperties: false
      properties:
        offer_id:
          type: string
        device_id:
          type: string
        site_id:

```

```

    type: string
  output_id:
    type: string
  available:
    type: boolean
  duration_seconds:
    type: integer
    minimum: 1
  max_power_w:
    type: number
    minimum: 0
    exclusiveMinimum: true
  price_usdc:
    type: string
    pattern: ^\d+\.\d{6}$
    description: Decimal USDC amount; six fractional digits, not base units.
  network:
    type: string
    description: CAIP-2 identifier, confirmed by deployment.
  terms_version:
    type: string
  measured_at:
    type: string
    format: date-time
required:
- offer_id
- device_id
- site_id
- output_id
- available
- duration_seconds
- max_power_w
- price_usdc
- network
- terms_version
- measured_at
QuoteRequest:
  type: object
  additionalProperties: false
  properties:
    offer_id:
      type: string
    client_nonce:
      type: string
    access_grant:
      type: string
      description: Opaque operator-issued permission for this site and output; never expose in
public
      logs.
    required:
      - offer_id
      - client_nonce
      - access_grant
    example:
      offer_id: offer_demo
      client_nonce: buyer_nonce_demo
      access_grant: operator_issued_grant
Quote:
  type: object
  additionalProperties: false
  properties:
    quote_id:
      type: string
    offer:
      $ref: '#/components/schemas/Offer'
    expires_at:
      type: string
      format: date-time
    start_by:
      type: string
      format: date-time
    request_digest:
      type: string
      description: Digest of the immutable authorized purchase context.
    refund_policy:
      type: string
      description: Versioned rule for failed or partial delivery; finalized before accepting
payment.
    required:
      - quote_id
      - offer
      - expires_at
      - start_by
      - request_digest
      - refund_policy
PurchaseRequest:

```

```

type: object
additionalProperties: false
properties:
  quote_id:
    type: string
  client_nonce:
    type: string
required:
- quote_id
- client_nonce
example:
  quote_id: quote_demo
  client_nonce: buyer_nonce_demo
AcceptedPurchase:
type: object
additionalProperties: false
properties:
  session:
    $ref: '#/components/schemas/Session'
  access_token:
    type: string
    description: Opaque bearer token for this session only; store securely and exclude from
receipts.
required:
- session
- access_token
Session:
type: object
additionalProperties: false
properties:
  session_id:
    type: string
  quote_id:
    type: string
  device_id:
    type: string
  state:
    type: string
    enum:
      - accepted
      - starting
      - running
      - completed
      - partially_completed
      - failed
  payment_state:
    type: string
    enum:
      - accepted
      - settlement_pending
      - settled
      - refund_pending
      - partially_refunded
      - refunded
  created_at:
    type: string
    format: date-time
  started_at:
    type: string
    format: date-time
  ended_at:
    type: string
    format: date-time
  delivered_seconds:
    type: integer
    minimum: 0
  energy_wh:
    type: number
    minimum: 0
  receipt_id:
    type: string
  failure_code:
    type: string
  payment_reference:
    type: string
    description: Application payment reference; not necessarily an onchain transaction hash.
required:
- session_id
- quote_id
- device_id
- state
- payment_state
- created_at
- delivered_seconds
- payment_reference
example:

```

```

    session_id: session_demo
    quote_id: quote_demo
    device_id: ecandle_demo
    state: accepted
    payment_state: accepted
    created_at: '2026-09-27T10:00:00Z'
    delivered_seconds: 0
    payment_reference: payment_demo
SignedMeasurement:
  type: object
  additionalProperties: false
  properties:
    device_id:
      type: string
    key_id:
      type: string
    payload_b64:
      type: string
      description: Base64 of exact device-signed bytes; decode and verify without reserializing.
    signature_b64:
      type: string
    signature_profile:
      type: string
      description: Versioned, allowlisted signature and payload interpretation profile.
    payload_digest:
      type: string
      description: 'sha256: followed by lowercase hex of decoded payload bytes.'
  required:
    - device_id
    - key_id
    - payload_b64
    - signature_b64
    - signature_profile
    - payload_digest
Receipt:
  type: object
  additionalProperties: false
  properties:
    receipt_id:
      type: string
    session_id:
      type: string
    quote_id:
      type: string
    request_digest:
      type: string
    client_nonce:
      type: string
    payment_reference:
      type: string
    settlement_tx_hash:
      type: string
      description: Optional; only populated after a real settlement transaction exists.
    session_state:
      type: string
      enum:
        - completed
        - partially_completed
        - failed
    delivered_seconds:
      type: integer
      minimum: 0
    energy_wh:
      type: number
      minimum: 0
    issued_at:
      type: string
      format: date-time
    measurements:
      type: array
      items:
        $ref: '#/components/schemas/SignedMeasurement'
    proof_state:
      type: string
      enum:
        - pending
        - ready
        - unavailable
    proof_bundle:
      $ref: '#/components/schemas/ProofBundle'
  required:
    - receipt_id
    - session_id
    - quote_id
    - request_digest
    - client_nonce

```

- payment\_reference
- session\_state
- delivered\_seconds
- issued\_at
- measurements
- proof\_state

ProofBundle:

- type: object
- additionalProperties: false
- properties:
  - profile:
    - type: string
    - description: Versioned HashAnchor verification profile; actual profile must be confirmed

before

- launch.
  - artifact\_b64:
    - type: string
    - description: Base64 of the complete portable proof artifact.
  - verifier\_version:
    - type: string

required:

- profile
- artifact\_b64
- verifier\_version

DeviceIdentity:

- type: object
- additionalProperties: false
- properties:
  - device\_id:
    - type: string
  - key\_id:
    - type: string
  - public\_key:
    - type: string
  - signature\_profile:
    - type: string
  - valid\_from:
    - type: string
    - format: date-time
  - status:
    - type: string
    - enum:
      - active
      - retired
      - revoked
  - revoked\_at:
    - type: string
    - format: date-time

required:

- device\_id
- key\_id
- public\_key
- signature\_profile
- valid\_from
- status

Error:

- type: object
- additionalProperties: false
- properties:
  - code:
    - type: string
  - message:
    - type: string
  - request\_id:
    - type: string
  - retryable:
    - type: boolean

required:

- code
- message
- request\_id
- retryable

example:

- code: QUOTE\_EXPIRED
- message: Request a new quote before making a new payment.
- request\_id: request\_demo
- retryable: false

Health:

- type: object
- additionalProperties: false
- properties:
  - status:
    - type: string
    - enum:
      - ok
      - degraded

```
    version:  
      type: string  
required:  
- status  
- version
```